

ME315_Lista3

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Configurando RStudio para lidar com SQLite

```
library(DBI)
library(RSQLite)

con <- dbConnect(SQLite(), "me315.sqlite")
```

```
SELECT * FROM PRESERVE
```

Table 1: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0

1.

```
SELECT DISTINCT STATE, FEE FROM PRESERVE
```

Table 2: 4 records

STATE	FEE
AZ	3
MT	0
AZ	0
MA	0

2.

```
SELECT DISTINCT DNO FROM EMPLOYEE
```

Table 3: 3 records

DNO
20
10
40

3.

```
SELECT DNO, SALARY FROM EMPLOYEE
```

Table 4: 6 records

DNO	SALARY
20	2000
10	2000
20	3000
40	500
10	400
20	9000

```
SELECT DISTINCT DNO, SALARY FROM EMPLOYEE
```

Table 5: 6 records

DNO	SALARY
20	2000
10	2000
20	3000
40	500
10	400
20	9000

```
SELECT DISTINCT DNO, SALARY FROM EMPLOYEE
ORDER BY DNO, SALARY
```

Table 6: 6 records

DNO	SALARY
10	400
10	2000
20	2000
20	3000
20	9000
40	500

4.

```
SELECT * FROM PRESERVE
WHERE STATE = 'MT' AND ACRES < 1000
```

Table 7: 2 records

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0

5.

```
SELECT * FROM PRESERVE
WHERE ACRES > 1200 AND ACRES <= 15000
```

Table 8: 1 records

PNO	PNAME	STATE	ACRES	FEE
2	PINE BUTTE SWAMP	MT	15000	0

6.

```
SELECT * FROM PRESERVE
WHERE STATE = 'MT' AND FEE = 0 AND ACRES > 10000
```

Table 9: 1 records

PNO	PNAME	STATE	ACRES	FEE
2	PINE BUTTE SWAMP	MT	15000	0

7.

```
SELECT * FROM PRESERVE
WHERE STATE = 'MT' OR STATE = 'MA'
```

Table 10: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
10	HOFT FARM	MA	90	0

8.

```
SELECT * FROM PRESERVE
WHERE STATE = 'MT' OR ACRES < 1000
```

Table 11: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0

9.

```
SELECT PNO, PNAME FROM PRESERVE
WHERE NOT FEE = 0
```

Table 12: 3 records

PNO	PNAME
5	HASSAYAMPA RIVER
80	RAMSEY CANYON
6	PAPAGONIA-SONOITA CREEK

10.

```
SELECT PNO, PNAME FROM PRESERVE
WHERE NOT FEE = 3 AND NOT FEE = 10
```

Table 13: Displaying records 1 - 10

PNO	PNAME
3	DANCING PRAIRIE
7	MULESHOE RANCH
40	SOUTH FORK MADISON
14	MCELWAIN-OLSEN
13	TATKON
9	DAVID H. SMITH

PNO	PNAME
11	MIACOMET MOORS
12	MOUNT PLANTAIN
1	COMERTOWN PRAIRIE
2	PINE BUTTE SWAMP

11.

```
SELECT * FROM PRESERVE
WHERE (STATE = 'AZ' AND FEE = 0) OR ACRES < 100
```

Table 14: 5 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
11	MIACOMET MOORS	MA	4	0
10	HOFT FARM	MA	90	0

12.

```
SELECT * FROM PRESERVE
WHERE (ACRES < 1000 AND FEE = 0) OR STATE = 'AZ'
```

Table 15: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
80	RAMSEY CANYON	AZ	380	3

13.

```
SELECT * FROM PRESERVE
WHERE NOT FEE > 0 OR (STATE = 'MT' AND ACRES > 1000)
```

Table 16: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0

14.

```
SELECT * FROM PRESERVE
WHERE NOT (STATE = 'MT' AND FEE = 0)
```

Table 17: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
80	RAMSEY CANYON	AZ	380	3
10	HOFT FARM	MA	90	0
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

15.

São equivalentes pois temos: $A \text{ and } (B \text{ or } C) = (A \text{ and } B) \text{ or } (A \text{ and } C)$

```
SELECT * FROM PRESERVE
WHERE STATE = 'MA' AND (ACRES > 1000 OR FEE = 0.0)
```

Table 18: 6 records

PNO	PNAME	STATE	ACRES	FEE
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
10	HOFT FARM	MA	90	0

```
SELECT * FROM PRESERVE
WHERE (STATE = 'MA' AND ACRES > 1000) OR (STATE = 'MA' AND FEE = 0.0)
```

Table 19: 6 records

PNO	PNAME	STATE	ACRES	FEE
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
10	HOFT FARM	MA	90	0

16.

São equivalentes pois temos: $A \text{ or } (B \text{ and } C) = (A \text{ or } B) \text{ and } (A \text{ or } C)$

```
SELECT * FROM PRESERVE
WHERE STATE = 'MA' OR (ACRES > 1000 AND FEE = 0.0)
```

Table 20: 9 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0

PNO	PNAME	STATE	ACRES	FEE
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
10	HOFT FARM	MA	90	0

```
SELECT * FROM PRESERVE
WHERE (STATE = 'MA' OR ACRES > 1000) AND (STATE = 'MA' OR FEE = 0.0)
```

Table 21: 9 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
10	HOFT FARM	MA	90	0

17.

Não são equivalentes pois temos: $\text{not (A and B)} = \text{not A or not B}$ dessa forma a primeira e a terceira cláusulas são equivalentes mas não são equivalentes a segunda cláusula

```
SELECT * FROM PRESERVE
WHERE NOT (ACRES < 50 AND STATE = 'MA')
```

Table 22: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0

PNO	PNAME	STATE	ACRES	FEE
14	MCELWAIN-OLSEN	MA	66	0
9	DAVID H. SMITH	MA	830	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3

```
SELECT * FROM PRESERVE
WHERE NOT ACRES < 50 AND NOT STATE = 'MA'
```

Table 23: 8 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

```
SELECT * FROM PRESERVE
WHERE ACRES >= 50 OR STATE <> 'MA'
```

Table 24: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
9	DAVID H. SMITH	MA	830	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3

18.

Não são equivalentes pois temos: $\text{not (A or B)} = \text{not A and not B}$ dessa forma a primeira e a terceira cláusulas são equivalentes mas não são equivalentes a segunda cláusula

```
SELECT * FROM PRESERVE
WHERE NOT (ACRES < 50 OR STATE = 'MA')
```

Table 25: 8 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

```
SELECT * FROM PRESERVE
WHERE NOT ACRES < 50 OR NOT STATE = 'MA'
```

Table 26: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
9	DAVID H. SMITH	MA	830	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3

```
SELECT * FROM PRESERVE
WHERE NOT ACRES < 50 AND NOT STATE = 'MA'
```

Table 27: 8 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3